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Antiatherosclerotic Properties of *Echinodorus grandiflorus* (Cham. & Schltdl.) Micheli: From Antioxidant and Lipid-Lowering Effects to an Anti-Inflammatory Role

Francielly Mourão Gasparotto,[1] Rhanany Alan Calloi Palozi,[1] Claudio Henrique Francisconi da Silva,[2] Karoline Bach Pauli,[2] Guilherme Donadel,[2] Bruno Henrique Lopes Botelho Lourenço,[2] Bruna Caroline Nunes,[2] Francislaine Aparecida dos Reis Lívero,[2] Lauro Mera de Souza,[3] Emerson Luiz Botelho Lourenço,[2] Cândida Aparecida Leite Kassuya,[1] and Arquimedes Gasparotto Junior[1]

[1]*Laboratory of Electrophysiology and Cardiovascular Pharmacology, Faculty of Health Sciences, Federal University of Grande Dourados (UFGD), Dourados, MS, Brazil.*
[2]*Laboratory of Preclinical Research of Natural Products, Paranaense University (UNIPAR), Umuarama, Paraná, Brazil.*
[3]*Institute of Research Pelé Pequeno Príncipe, Pequeno Príncipe Faculty, Curitiba, Paraná, Brazil.*

ABSTRACT *Echinodorus grandiflorus* is an important medicinal plant species that is native to South America. Despite extensive popular usage as a hypolipidemic drug, its effects as an atheroprotective agent remain unknown. The aim of this study was to evaluate the effects of an ethanol-soluble fraction that was obtained from *E. grandiflorus* (ESEG) leaves against the development of atherosclerosis in rabbits. Male rabbits received a diet that was supplemented with 1% cholesterol (cholesterol-rich diet [CRD]) for 60 days. After 30 days of the CRD, the animals were divided into five groups ($n = 6$) and treated with ESEG (10, 30, and 100 mg/kg), simvastatin (2.5 mg/kg), or vehicle once daily for 30 days. The negative control group was fed a cholesterol-free diet and treated orally with vehicle. At the end of 60 days, serum lipids, oxidized low-density lipoprotein, thiobarbituric acid reactive substances, nitrotyrosine, and serum interleukin 1 beta (IL-1β), IL-6, soluble intercellular adhesion molecule-1 (sICAM-1), and soluble vascular cell adhesion molecule-1 (sVCAM-1) levels were determined. Samples from the aortic arch and thoracic segment were also collected to investigate the tissue antioxidant defense system and perform histopathological analysis. Oral ESEG administration significantly reduced serum lipid levels in CRD-fed rabbits. This treatment also modulated the arterial antioxidant defense system by reducing lipid and protein oxidation. Similarly, serum IL-1β, IL-6, sICAM-1, and sVCAM-1 levels significantly decreased, accompanied by a reduction of atherosclerotic lesions in all arterial branches. These findings suggest that ESEG may be a new herbal medicine that can be directly applied for the treatment and prevention of atherosclerotic disease.

KEYWORDS: • ***alismataceae*** • ***anti-inflammatory*** • ***antioxidant*** • ***atherosclerosis*** • ***dyslipidemia***

INTRODUCTION

CARDIOVASCULAR DISEASES, especially atherosclerosis, exert an important influence on patients' quality of life, being considered one of the most important reasons for physical disability and premature death around the world. It is characterized by a silent and progressive onset, accompanied by aggression to the arterial surface, a result of an inflammatory and fibroproliferative response of the intima layers of the arteries.[1] The initial event of atherogenesis occurs through vascular endothelial function disorder, and the main contributory factor for triggering is hypercholesterolemia.[2] The lesion caused by the presence of the atherosclerotic plaque may compromise several arterial segments and is responsible for several adjacent complications such as coronary artery disease, intermittent limb claudication, stroke, and abdominal aortic aneurysm.[3]

Primary prevention of dyslipidemia and atherosclerosis should be made with lifestyle changes. Although effective in most cases of nonfamilial hypercholesterolemia, it is known that the effectiveness of this type of intervention is variable and depends on the patient's adherence, type of diet adopted, regular physical activity, as well as the cessation of smoking and alcoholism.[2] Although lifestyle changes are quite effective, in some cases pharmacological treatment is necessary.

Currently there is a vast therapeutic arsenal for the treatment of dyslipidemias and for the prevention of atherosclerosis. These include cholesterol absorption inhibitors

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Address correspondence to: Arquimedes Gasparotto Junior, PhD, Laboratory of Electrophysiology and Cardiovascular Pharmacology, Faculty of Health Sciences, Federal University of Grande Dourados, Rodovia Dourados-Itahum, km 12, P.O. Box 533, 79.804-970 Dourados, MS, Brazil, E-mail: arquimedesgasparotto@gmail.com

(*e.g.*, ezetimibe), bile acid sequestrants (*e.g.*, cholestyramine and colestipol), niacin and its derivatives (*e.g.*, nicofuranose and niceritrol), fibrates (*e.g.*, clofibrate and ciprofibrate), cholesteryl ester transfer protein (CETP) inhibitors (*e.g.*, anacetrapib and torcetrapib), and statins (*e.g.*, simvastatin, atorvastatin, and rosuvastatin).[4]

Despite the efficacy of statins, the appearance of adverse effects such as changes in hepatic function, myalgia, and to a lesser extent rhabdomyolysis, has stimulated the research of several alternative treatments, including medicinal plants.[5] Thus, there is currently great interest in the search for alternative therapeutic options for the prevention and treatment of dyslipidemias, especially to be used concomitantly with conventional therapies.[6,7] In this regard, a well-accepted option comes from natural products, especially those that are commonly used by the population and are part of the cultural arsenal transmitted by generations.

Echinodorus grandiflorus (Cham. & Schltdl.) Micheli. (Alismataceae), popularly known as ''chapéu de couro,'' ''chá-de-campanha,'' and ''erva do brejo,'' is a plant species that is native to South America. The infusion of its leaves has been used as an antihypertensive and diuretic agent by different native populations.[8,9] Because of its extensive ethnobotanical use in Brazil, the genus *Echinodorus* was included as a hypolipidemic and diuretic agent in the fifth edition of the Brazilian Pharmacopoeia.[10,11]

Several preclinical studies have reported that *E. grandiflorus* is a promising species for the treatment of cardiovascular diseases. Different preparations that were obtained from this species were shown to have diuretic,[12,13] anti-edematous,[14] antihypertensive,[13,15] and vasodilatory[16] properties. Surprisingly, despite extensive popular usage, its hypolipidemic and antiatherosclerotic effects remain unknown. This study employed a classic model of atherosclerosis in male New Zealand rabbits to investigate the effects of *E. grandiflorus* against dyslipidemia and atherosclerosis. We also evaluated the possible molecular pathways that may be involved in its cardioprotective actions.

MATERIALS AND METHODS

Drugs and reagents

The following drugs, salts, and solutions were used: isoflurane and potassium chloride (Cristália, Itapira, SP, Brazil). Simvastatin and cholesterol were obtained from Sigma-Aldrich (St. Louis, MO, USA). All of the other reagents were obtained in analytical grade.

Plant material and preparation of the ethanol-soluble fraction

E. grandiflorus leaves (5.0 kg) were collected in February 2014 from the botanical garden of the Universidade Paranaense (UNIPAR, Umuarama, Brazil) at 430 m above sea level (S23°47′55–W53°18′48). A voucher specimen was deposited in the Herbarium of the Universidade Estadual de Maringá (HUEM, no. 20810). From the dried and crushed leaves of *E. grandiflorus* we obtained the ESEG (yield: 9.54% w/w). The procedure to obtain ESEG and the detailed phytochemical analysis were recently published by Gasparotto *et al.*[17]

Animals

Fourteen-week-old male New Zealand rabbits, weighing 1.8–2.0 kg, were obtained from Universidade Federal do Paraná (UFPR, Brazil) and housed at the UNIPAR vivarium under controlled temperature (20 ± 2°C), humidity (50 ± 10%), and a 12/12 h light/dark cycle with *ad libitum* access to food and water. All of the experimental procedures were approved by the Institutional Ethics Committee of Universidade Federal da Grande Dourados (UFGD, Dourados, Brazil; protocol no. 08/2015) and conducted in accordance with the Brazilian Legal Framework on the Scientific Use of Animals.

Experimental procedures

Dyslipidemia and atherosclerosis were induced according to Barboza *et al.*[18] The rabbits received *ad libitum* a commercial diet (Purina® Rabbit Chow®, St. Louis, Missouri, EUA) that was supplemented with 1% cholesterol (cholesterol-rich diet [CRD]) for 60 days. After 30 days, the animals were randomly distributed into five groups ($n = 6$/group) and orally treated by gavage with ESEG (10, 30, and 100 mg/kg), vehicle (filtered water, 1 mL/kg; positive control), or simvastatin (2.5 mg/kg) daily for 30 days. One group was fed a cholesterol-free diet and was treated only with vehicle (negative control).

Body weight was measured at the beginning of the experiments and after 30 and 60 days. The water and feed intake was also monitored weekly. On day 60, all of the animals were fasted for 8 h and anesthetized with isoflurane. Blood samples were then collected from the jugular vein. Serum was obtained by centrifugation (1500 *g* for 5 min).

Triglyceride (TG), total cholesterol (TC), and high-density lipoprotein cholesterol (HDL-C) levels were measured using an automated biochemical analyzer (Roche Cobas Integra 400 plus). Serum very low-density lipoprotein cholesterol/LDL cholesterol (VLDL/LDL-C) levels were calculated as the difference between TC and HDL-C. Serum oxidized LDL (ox-LDL), nitrotyrosine (NT), soluble vascular cell adhesion molecule-1 (sVCAM-1), soluble intercellular adhesion molecule-1 (sICAM-1), interleukin-1β (IL-1β), and IL-6 levels were measured by enzyme-linked immunosorbent assay (ELISA; BD Biosciences, San Jose, CA, USA). Malondialdehyde (MDA) levels were measured using an MDA assay kit (Cayman Chemical, Ann Arbor, MI, USA).

After blood collection, the rabbits were euthanized by an intravenous injection of potassium chloride. Aorta segments (aortic arch, thoracic, abdominal, and aortoiliac segments) were removed and fixed in 10% formalin. The aortic arch was considered from the exteriorization of the aorta in the heart to the bifurcation of the left subclavian artery. The thoracic segment was considered from the end of the bifurcation of the left subclavian artery to the diaphragm; and the abdominal portion comprised between the diaphragm and the bifurcation of the iliac arteries. Finally, the aortoiliac segments were obtained from the bifurcation of the left iliac artery to the inguinal ligament.

After 48 h, a part of each arterial branch was longitudinally sectioned, rinsed in 70% alcohol, and immersed in Sudan-IV staining solution at room temperature for 15 min. The tissues were transferred to 80% alcohol for 20 min and washed in tap water for 1 h, and the luminal surface was assessed for sudanophilic lesions. The analysis included digital image capture by Motic Images Plus 2.0 software, image processing for edge detection (lesion identification), and, finally, computation of lesions. Diseased portions of the vessels stained red with Sudan IV dye, appeared black or dark gray on the computer monitor. The edited image was separated into diseased and nondiseased areas by using an iterative algorithm for multiple threshold detection. Finally, the program provided the area corresponding to the lesions that were subtracted from the total area of the vessel.

A second part of the arterial branches was dehydrated with alcohol and xylene, embedded in paraffin, sectioned at 5 μm, stained with hematoxylin/eosin, and microscopically examined. The intima and media layers were measured using a caliper. For each animal, we assemble several blades with the entire arterial branch studied. All neointimal formation (per slide) was identified, and for each atherosclerotic formation six lines of division were established. For each neointimal formation were performed six measurements, and one mean for each slide was obtained. When all the slides were evaluated, a new mean for each animal was determined. The area of the microscope was adjusted using an ocular micrometer coupled with the 100× objective, determining the size of the field per square micrometer. All images were obtained and evaluated by Motic Images Plus 2.0 software.

Finally, a third part of the arterial branches was sectioned and homogenized in K^+ phosphate buffer (0.1 M, pH 6.5). Superoxide dismutase (SOD) was analyzed using the pyrogallol oxidation method.[19] Lipid peroxidation (LPO) rate was measured by the ferrous oxidation-xylenol orange method according to Jiang *et al.*[20] Glutathione (GSH) levels were measured based on a previously described technique,[21] with a few modifications. The amount of protein in the homogenates was determined using the Bradford method.[22]

Statistical analysis

The results are expressed as the mean ± standard deviation of six animals per group. The statistical analyses were performed using one-way analysis of variance (ANOVA) followed by Bonferroni's test or the Kruskal–Wallis test followed by Dunn's test. Values of $P < .05$ were considered statistically significant. The graphs were drawn and the statistical analysis was performed using Prism 5.0 software (GraphPad, San Diego, CA, USA).

RESULTS

ESEG treatment prevents progressive weight loss in CRD-fed rabbits

Body weight gain in rabbits that were fed the CRD and treated with ESEG, simvastatin, or vehicle is shown in Figure 1. Negative control animals had a body weight of 2.38 ± 0.52 kg at the beginning of the experiments, which increased to 3.44 ± 0.77 and 3.68 ± 0.51 kg after 30 and 60 days, respectively. Positive control rabbits had significantly lower body weight gain after 30 days (2.89 ± 0.46 kg) and 60 days (2.92 ± 0.35 kg). Similarly, animals that were treated with ESEG (10, 30, or 100 mg/kg) or simvastatin had body weight gain that was similar to positive control animals at the end of 30 days.

At the end of the experimental period, all of the animals that were treated with ESEG (30 and 100 mg/kg) had body weight gain that was similar to rabbits that were treated with the standard commercial diet, with a body weight profile that was significantly different from animals that were fed the CRD and treated only with vehicle. Moreover, the changes in body weight were related to the feed consumption. Animals fed with standard commercial diet (negative control) maintained a regular food intake. In contrast, the groups fed with CRD started to consume less feed during the experiment. Interestingly, this change was reversed after ESEG treatment at doses of 30 and 100 mg/kg (data not shown).

Prolonged treatment with ESEG significantly reduces serum lipids in CRD-fed rabbits

The TC, HDL-C, VLDL/LDL-C, and TG levels in negative control animals were 64 ± 12, 16 ± 1.8, 49 ± 12, and 121 ± 8.8 mg/dL, respectively. Positive control rabbits

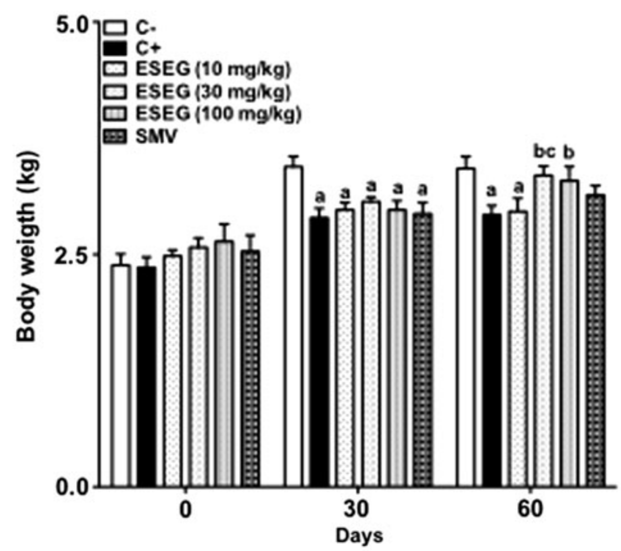

FIG. 1. ESEG treatment prevents progressive weight loss in CRD-fed rabbits. Body weight was measured at the beginning of experiments (day 0) and after 30 and 60 days of treatment with ESEG (10, 30, and 100 mg/kg), SMV (2.5 mg/kg), or vehicle (1 mL/kg). Values are expressed as mean ± SD ($n = 6$) in comparison with C^- ($^aP < .05$), C^+ ($^bP < .05$), or previous ESEG dose ($^cP < .05$) using one-way ANOVA followed by Bonferroni's test. ANOVA, analysis of variance; C^-, negative control group; C^+, positive control group; CRD, cholesterol-rich diet; ESEG, ethanol-soluble fraction obtained from *E. grandiflorus*; SD, standard deviation; SMV, simvastatin.

exhibited increases in serum lipid levels (1739 ± 125, 35 ± 3.6, 1705 ± 125, and 211 ± 18 mg/dL for TC, HDL-C, VLDL/LDL-C, and TG, respectively; Fig. 3A–D). Oral ESEG administration for 30 days dose-dependently reduced TC, VLDL/LDL-C, and TG levels, reaching to 710 ± 130, 673 ± 129, and 46 ± 8.1 mg/dL, respectively, at the dose of 100 mg/kg. At the highest dose, ESEG induced a lipid-lowering effect that was similar to simvastatin treatment. The CRD increased HDL-C levels in all of the experimental groups, and HDL-C levels remained unchanged regardless of treatment with ESEG or simvastatin (Fig. 2A–D).

ESEG treatment reduces macroscopic lesions in aorta segments from rabbits undergoing CRD

Atherosclerotic lesions in aorta segments were used as an index of disease severity (Fig. 3A–E). The total average area of the aortic arch, thoracic, abdominal, and aortoiliac branches were 63 ± 6, 550 ± 35, 228 ± 22, and 76 ± 8 mm^2, respectively. The area percentage of sudanophilic lesions in the aortic arch, thoracic, abdominal, and aortoiliac segments in the positive control group after 60 days of the CRD were 35 ± 8.1%, 9.2 ± 1.8%, 12 ± 2.0%, and 17 ± 2.0%, respectively (Fig. 3A–D).

The area percentage of sudanophilic lesions in the aortic arch, thoracic, abdominal, and aortoiliac segments in animals that were treated with ESEG at 100 mg/kg decreased to 16 ± 2.5%, 4.0 ± 0.8%, 4.7 ± 1.7%, and 5.7 ± 0.9%, respectively (Fig. 3A–D). ESEG at the dose of 30 mg/kg reduced sudanophilic lesions in the aortic arch (20 ± 6.1%) and abdominal segment (7.5 ± 1.4%). Although simvastatin decreased the areas of sudanophilic lesions in all arterial branches, the results were statistically lower than treatment with 100 mg/kg ESEG.

Prolonged treatment with ESEG reduces atherosclerotic lesions in the tunica intima from aorta segments of CRD-fed rabbits

Morphometric measures of the intima layer of aortic arch, thoracic, abdominal, and aortoiliac segments are shown in Figure 4A–D. The histopathological examination of aorta segments in the positive control group showed significant thickness in the intima layer (aortic arch: 28 ± 7.8 μm; thoracic segment: 30 ± 8.7 μm; abdominal segment: 52 ± 12 μm; aortoiliac segment: 36 ± 9.4 μm) compared with the negative control group (aortic arch: 1.95 ± 0.72 μm; thoracic segment: 2.02 ± 0.51 μm; abdominal segment: 1.80 ± 0.45 μm; aortoiliac segment: 1.79 ± 0.60 μm). Treatment with ESEG (30 and 100 mg/kg) reduced the thickness of the intima layer in all CRD-fed rabbits, with values that were very similar to animals that were treated with simvastatin. No significant morphometric differences in the media layers were observed among groups (data not shown).

ESEG reduces serum levels of oxidative/nitrosative stress markers and modulates the arterial antioxidant defense system of CRD-fed rabbits

Positive control animals exhibited significant increases in thiobarbituric acid reactive species (TBARS) and NT levels (~75% increase) compared with negative control animals (Fig. 5A–C). Similarly, ox-LDL levels increased from 75 ± 12.9 mU/mL in the negative control group to 432 ± 70 mU/mL in rabbits that were fed the CRD. Treatment with 30 and 100 mg/kg ESEG reduced TBARS and NT levels to values that were close to negative control animals (negative controls: 2.53 ± 0.41 M TBARS, 0.013 ± 0.002 μM NT; 30 mg/kg ESEG: 3.07 ± 0.74 M TBARS, 0.016 ± 0.001 μM NT; 100 mg/kg ESEG: 2.31 ± 0.62 M TBARS, 0.012 ± 0.002 μM NT).

Similarly, ox-LDL levels were significantly reduced by ESEG (30 and 100 mg/kg), with values that were close to negative control animals at the highest dose (30 mg/kg ESEG: 257 ± 74 mU/mL; 100 mg/kg ESEG: 114 ± 45 mU/mL). Although animals that were treated with simvastatin also presented an interesting antioxidant response, the values were less expressive than those in animals that were treated with ESEG (Fig. 5A–C).

The results of the analysis of the arterial antioxidant defense system are presented in Table 1. The CRD increased LPO levels by ~168% in positive control animals. Treatment with 100 mg/kg ESEG reversed this increase in all of the evaluated tissues. Treatment with 100 mg/kg ESEG also reversed the decreases in SOD activity and GSH levels that were observed in positive control animals.

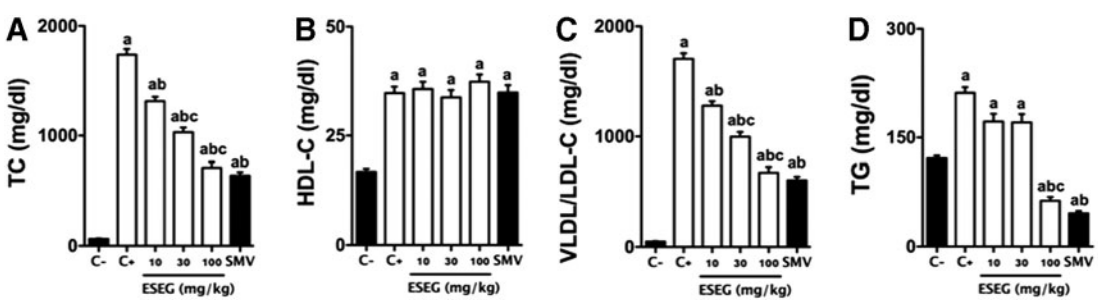

FIG. 2. Prolonged ESEG treatment reduces lipid serum levels of CRD-fed rabbits. Serum sample were obtained on the 60th day of experiments and TC **(A)**, HDL-C **(B)**, VLDL/LDL-C **(C)**, and TG **(D)** levels were analyzed. Values are expressed as mean ± SD (n = 6) in comparison with C$^-$ (^aP < .05), C$^+$ (^bP < .05) or previous ESEG dose (^cP < .05) using one-way ANOVA followed by Bonferroni's test. HDL-C, high-density lipoprotein cholesterol; TC, total cholesterol; TG, triglyceride; VLDL/LDL-C, very low-density lipoprotein/LDL cholesterol.

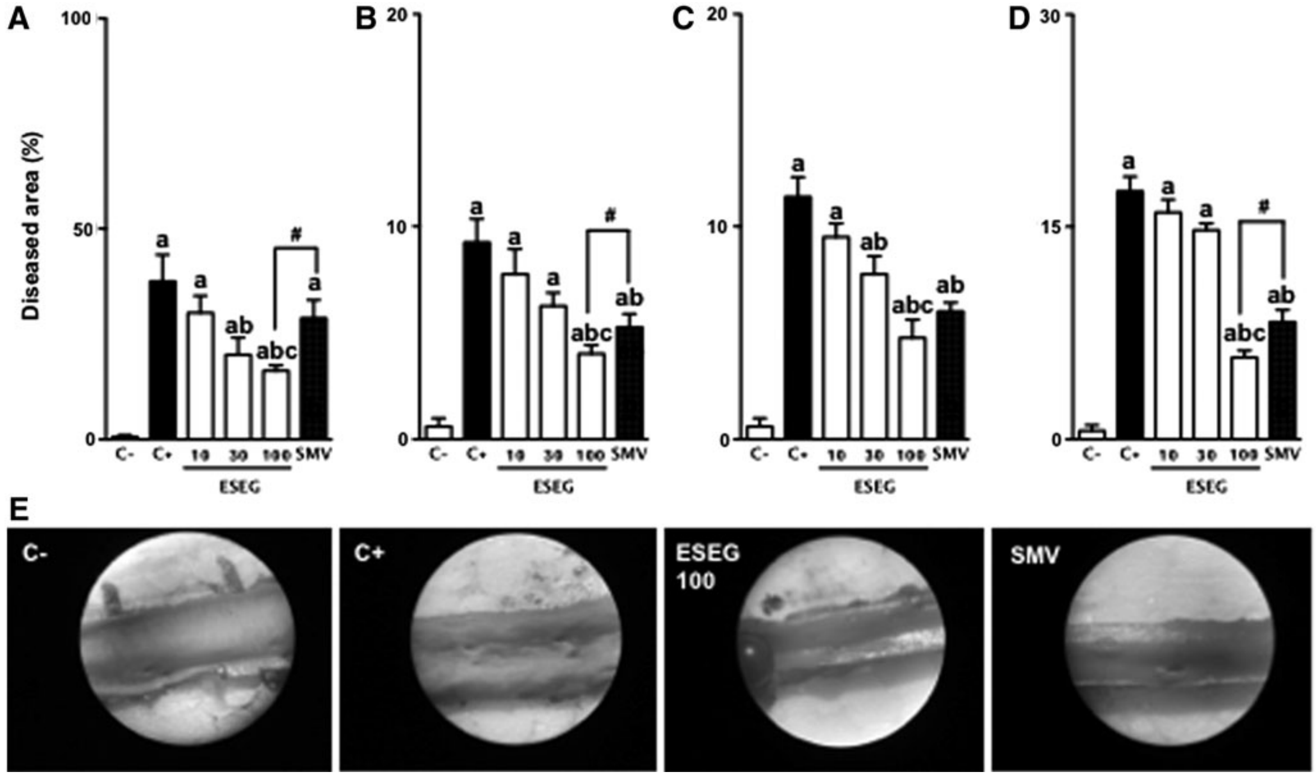

FIG. 3. ESEG treatment reduces sudanophilic lesions in aorta segments from rabbits undergoing CRD. Percentage of atherosclerotic lesions in the aortic arch **(A)**, thoracic **(B)**, abdominal **(C)**, and aortoiliac segments **(D)**, and representative macroscopic observations in thoracic segments **(E)** are shown. Results are expressed as percentage of the area covered by plaque over the total area evaluated. Images in 4×scale. Values are expressed as mean ± SD ($n=6$) in comparison with C^- ($^aP<.05$), C^+ ($^bP<.05$), SMV ($^\#P<.05$), or previous ESEG dose ($^cP<0.05$) using one-way ANOVA followed by Bonferroni's test.

ESEG treatment induces important anti-inflammatory response in atherosclerotic rabbits

Serum IL-1β, IL-6, sICAM-1, and sVCAM-1 levels in CRD-fed rabbits that were treated with ESEG and simvastatin are shown in Figure 6A–D. IL-1β, IL-6, sICAM-1, and sVCAM-1 levels in negative control animals (baseline: 483 ± 90 pg/mL, 214 ± 39 ng/L, 3.92 ± 0.65 ng/L, and 1.60 ± 0.55 ng/L, respectively) increased to 884 ± 92 pg/mL, 542 ± 91 ng/L, 11.89 ± 1.87 ng/L, and 4.95 ± 0.77 ng/L, respectively, in positive control animals. In CRD-fed rabbits, treatment with 100 mg/kg ESEG reduced IL-1β, IL-6, sICAM-1, and sVCAM-1 levels to values that were close to negative control animals and simvastatin-treated animals.

DISCUSSION

One of the main triggers of the atherosclerotic process is a CRD. Excess dietary cholesterol, mainly LDL-C and VLDL-C, when not properly transported and metabolized, remains in the systemic circulation in large amounts, thus contributing directly to the development of endothelial dysfunction and atherosclerosis.[23] Moreover, the CRD can also increase vascular oxidative stress, inducing oxidative/nitrosative imbalance and leading to LDL-C oxidation, endothelial cell activation, monocyte recruitment, and foam cell formation.[24]

In this study we induced significant dyslipidemia in New Zealand rabbits by administering CRD. We also observed an oxidative/nitrosative imbalance that led to a significant increase in ox-LDL levels. As a result, we found an expressive increase in atherosclerotic plaque formation in different arterial branches. In fact, rabbits constitute excellent models for the study of the development of atherosclerosis. They share important characteristics with humans that are not observed in rodents, including high sensitivity to dietary cholesterol, hepatic LDL-C receptor, CETP, presence of the VLDL-C receptor on macrophages, and heterogeneous HDL-C particles.[25]

The use of synthetic lipid-lowering drugs—including statins—have shown significant efficacy in reducing cholesterol levels in different patients. It is now known that the mechanism of action of statins to obtain cholesterol reduction is due to the inhibition of the enzyme 3-Hydroxy-3-methylglutaryl coenzyme (HMG-CoA) reductase, through an affinity of these drugs with the active site of the enzyme. This inhibition is reversible and competitive with the HMG-CoA substrate.[26] Furthermore, part of the cardioprotective effect of statins derives from complex pleiotropic effects, including antioxidant and anti-inflammatory activity.[27]

Despite the efficacy of statins, the appearance of side effects such as changes in hepatic function and myalgia has stimulated the research of several alternative treatments,

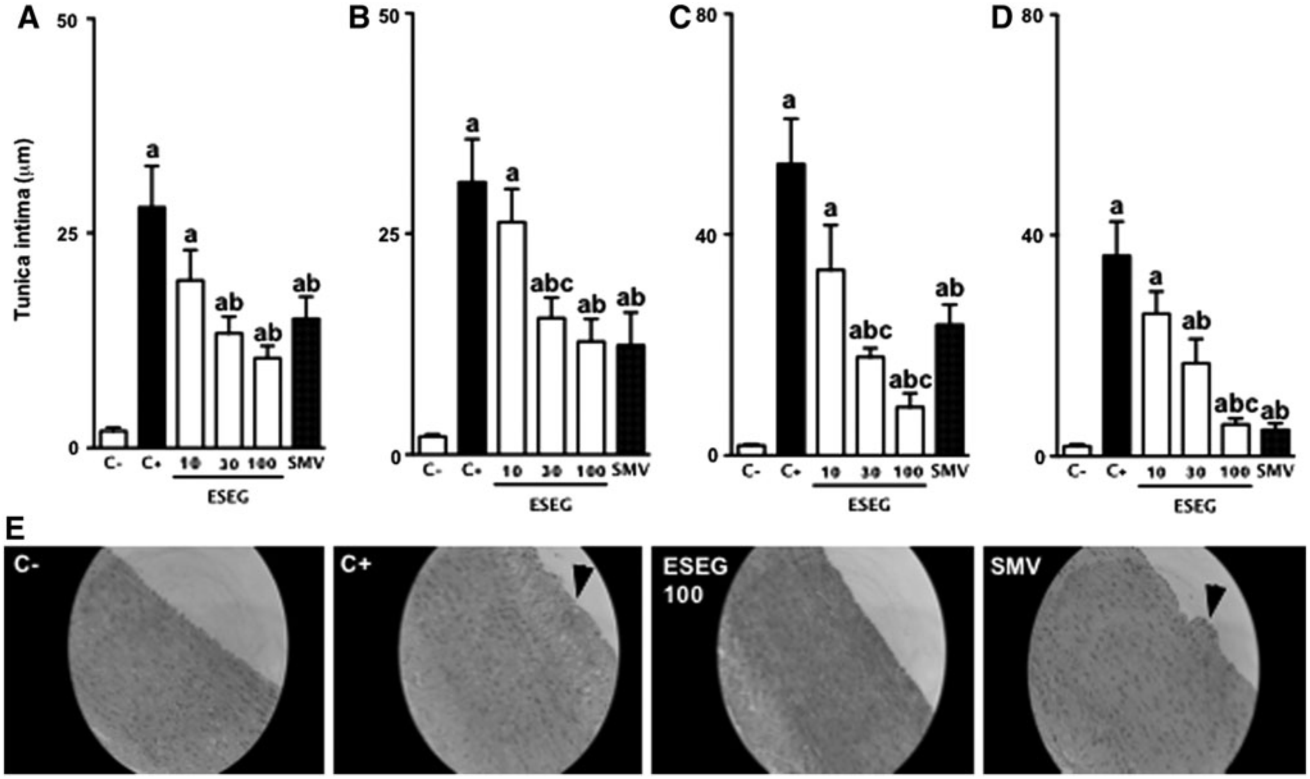

FIG. 4. ESEG treatment reduces arterial thickness in aorta segments from rabbits undergoing CRD. Histomorphometric analyses of tunica intima from aortic arch **(A)**, thoracic **(B)**, abdominal **(C)**, and aortoiliac segments **(D)** are shown. Representative cross sections of the abdominal segment stained with hematoxylin-eosin are also presented **(E)**. *Black arrows* show neointimal formation. Images in 40× scale. Values are expressed as mean ± SD ($n=6$) in comparison with C^- ($^aP<.05$), C^+ ($^bP<.05$), SMV ($^\#P<.05$), or previous ESEG dose ($^cP<.05$) using one-way ANOVA followed by Bonferroni's test.

including herbal medicines.[28] Although herbal medicines are not completely free of side effects, many species commonly used by the population have shown to be safe and highly effective in reducing plasma lipids.[29] Most often the available studies do not precisely define the mechanisms involved in the lipid-lowering properties of medicinal species. However, different secondary metabolites, including saponins, tannins, alkaloids, and/or flavonoids, can act synergistically, affecting the absorption, metabolism, or excretion of cholesterol.[30]

E. grandiflorus is frequently used in Brazil as a lipid-lowering agent.[10,11] A recent study by Gasparotto *et al.*[17]

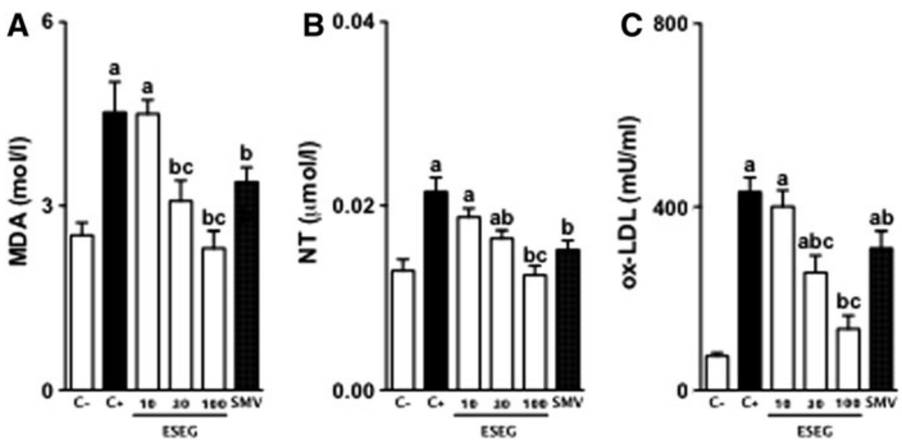

FIG. 5. ESEG reduces serum levels of oxidative/nitrosative stress markers. Serum MDA **(A)**, NT **(B)**, and ox-LDL levels **(C)** are presented. Values are expressed as mean ± SD ($n=6$) in comparison with C^- ($^aP<.05$), C^+ ($^bP<.05$), SMV ($^\#P<.05$), or previous ESEG dose ($^cP<.05$) using one-way ANOVA followed by Bonferroni's test. MDA, malondialdehyde; NT, nitrotyrosine; ox-LDL, oxidized LDL.

Table 1. Effects of Oral Administration of Ethanol-Soluble Fraction Obtained from *E. grandiflorus* on Antioxidant Tissue Defense System

Parameter	C^-	C^+	*ESEG (10 mg/kg)*	*ESEG (30 mg/kg)*	*ESEG (100 mg/kg)*	*SMV (2.5 mg/kg)*
SOD	83.83 ± 6.21	26.34 ± 4.22[a]	42.70 ± 12.1	65.18 ± 9.10	73.53 ± 2.99[b]	35.11 ± 9.11[a]
GSH	373.20 ± 13.2	126.50 ± 9.12[a]	217.90 ± 61.35	349.50 ± 29.11	402.10 ± 44.12[bc]	232.20 ± 71.22
LPO	6.24 ± 1.85	16.74 ± 2.01[a]	8.34 ± 0.92	7.60 ± 1.82	6.01 ± 1.82[b]	11.32 ± 1.99

Values are expressed as mean ± standard deviation of six rabbits in each group in comparison with C^- ($^aP < .05$), C^+ ($^bP < .05$) or ESEG previous dose ($^cP < .05$) using one-way Kruskal–Wallis followed by Dunn's test.

C^-, negative control group; C^+, positive control group; ESEG, ethanol-soluble fraction obtained from *E. grandiflorus*; GSH, reduced glutathione (μg GSH/g of tissue); LPO, lipid peroxidation (nmol hydroperoxides/mg of protein); SMV, simvastatin; SOD, superoxide dismutase (unit of SOD/mg of protein).

using the same experimental model showed that the CRD was able to induce significant cardiac alterations in male rabbits. Myocardial flaccidity, fatty degeneration, and concentric left ventricular hypertrophy were observed in all CRD-fed rabbits. Treatment with ESEG, especially the highest dose, significantly reduced all cardiac changes that are induced by CRD.

So, we hypothesized that ESEG may also have atheroprotective effects. Confirming our hypothesis, 30-day treatment with ESEG reduced serum lipid levels, especially at higher doses. A modulatory effect on the arterial antioxidant defense system was also observed, with a significant antioxidant and antinitrosant response. Consequently, atherosclerotic lesions significantly decreased, and effects that were superior to simvastatin were also observed in some arterial branches.

When considering lipid-lowering effects, simvastatin presented a response that was very similar to ESEG. However, ESEG appeared to have higher antioxidant potential, reflected by a greater preventive effect on the formation of atherosclerotic lesions. Our results allow us to conjecture that ESEG has additional cardioprotective effects to statins because, in addition to effectively reducing serum lipids, it has shown important antioxidant, diuretic, and hypotensive effects.[12,13]

Since statins have well-defined pleiotropic effects, we chose to evaluate whether ESEG would also be able to have these activities. To investigate the pleiotropic effects of ESEG, we measured the levels of inflammatory markers that commonly increase during the atherosclerotic process.

Endothelium damage that is caused by the accumulation of ox-LDL-C can promote the release of various inflammatory mediators, including IL-1β and IL-6, and induce the expression of different adhesion molecules, such as sICAM-1 and sVCAM.[31] These adhesion molecules are essential for the adhesion, rolling, and migration of leukocytes from the bloodstream to the site of injury.[32] Thus, monocytes are able to transmigrate into the subendothelial space where, after phagocytizing ox-LDL and metamorphose into foam cells, they progressively accumulate to form atherosclerotic plaques.[33]

In this study, CRD-fed rabbits that were treated only with vehicle exhibited significant increases in IL-1β, IL-6, sICAM-1, and sVCAM levels and consequently an increase in the thickness of the arterial intima layer. All of the animals that were treated with ESEG or simvastatin exhibited reductions of the levels of inflammatory markers, directly contributing to atheroprotective effects. The atheroprotective effects of ESEG, in addition to its lipid-lowering effects, appear to derive from a significant reduction of lipid

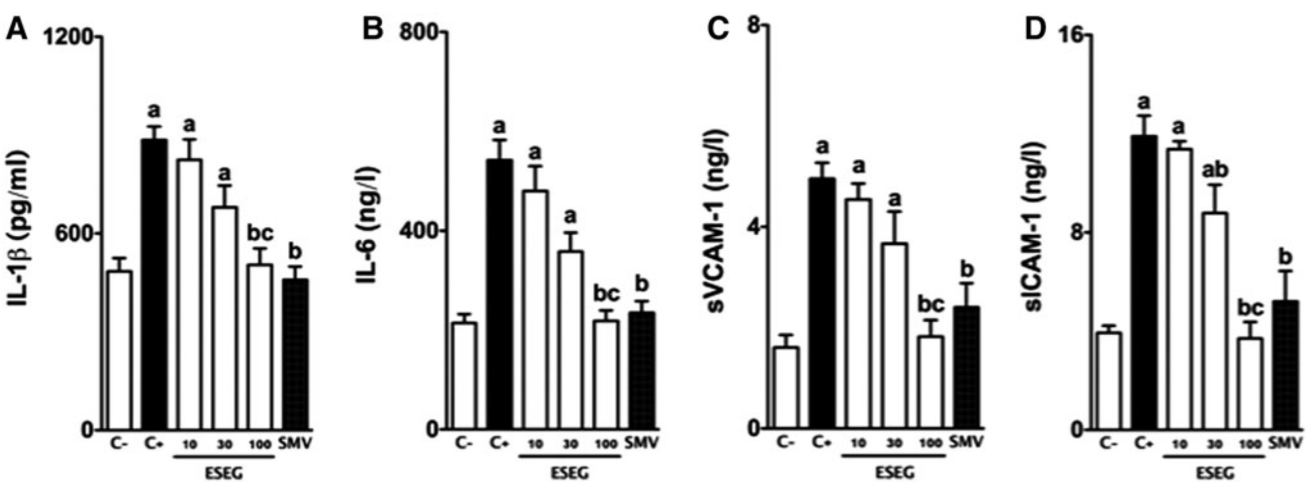

FIG. 6. ESEG treatment reduces ILs and soluble adhesion molecules levels in CRD-fed rabbits. Serum IL-1β (**A**), IL-6 (**B**), sVCAM-1 (**C**), and sICAM-1 (**D**) concentration are shown. Values are expressed as mean ± SD ($n = 6$) in comparison with C^- ($^aP < .05$), C^+ ($^bP < .05$), or previous ESEG dose ($^cP < .05$) using one-way ANOVA followed by Bonferroni's test. IL-1β, interleukin 1 beta; sICAM-1, soluble intercellular adhesion molecule-1; sVCAM-1, soluble vascular cell adhesion molecule-1.

oxidation, which possibly decreases endothelial damage by reducing IL-1β, IL-6, sICAM-1, and sVCAM-1 expression and monocyte transmigration through the vascular wall.

The main chemical constituents that are present in this species have been characterized. Several diterpenoids, phenolic acids, flavonols, alkaloids, saponins, and tannins have been identified.[14] Moreover, large amounts of phenolic compounds, mainly C-glycoside flavonoids (*e.g.*, isoorientin, isoorientin-*O*-rhamnoside, isoorientin-*O*-rhamnoside-dimethylether, isoorientin 7,3′-dimethylether, swertiajaponin, swertiajaponin-*O*-rhamnoside, isovitexin, isovitexin-*O*-rhamnoside, swertisin, and swertisin-*O*-rhamnoside) have been identified.[12,13]

In a recent study we have shown that the ESEG has abundant amounts of C-glycoside flavonoids, including isoorientin, swertiajaponin, isovitexin, and swertisin derivatives.[17] Previous studies reported that isovitexin (apigenin-6-C-glucoside), an isomer of vitexin, generally purified together with vitexin has an important antioxidant, anti-inflammatory, and vasodilatory effect.[34,35] Similarly, other flavonoids such as quercetin and its glycosylated derivatives showed inhibitory effects on the oxidative modification of LDL by macrophages, and naringenin 7-O-cetyl ether showed a significant inhibitory effect of HMG-CoA reductase.[36,37] In addition, several studies have shown that flavonoid supplementation prevents hepatic steatosis, dyslipidemia, and insulin resistance primarily through inhibition of hepatic fatty acid synthesis and increased fatty acid oxidation.[7,38]

Therefore, the present data suggest that the lipid lowering, antioxidant, anti-inflammatory, and antiatherogenic effects of ESEG may be attributable to coordinated actions of its secondary metabolites, mainly to flavonoids and their glycosylated derivatives.

In conclusion, ESEG possesses secondary metabolites that are responsible for significant hypolipidemic, antioxidant, and antinitrosant properties, which can modulate the local inflammatory process by reducing the evolution of atherosclerotic disease. The present results suggest that ESEG may be a new herbal medicine that can be directly applied for the treatment and prevention of atherosclerotic disease.

AUTHORS' CONTRIBUTIONS

All of the authors participated in the design and interpretation of the study, analysis of the data, and review of the article. F.M.G., R.A.C.P., C.H.F.S., F.A.R.L., K.B.P., G.D., B.H.L.B.L., B.C.N., L.M.S., and E.L.B.L. conducted the experiments; F.M.G., C.A.L.K., A.G.J., and F.A.R.L. were responsible for data discussion and article revision. A.G.J. was the senior researcher who was responsible for this study. All of the authors read and approved the final article.

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ETHICAL STATEMENT

All applicable international, national, and/or institutional guidelines for the care and use of animals were followed. All of the procedures that involved animals were in accordance with the ethical standards of the institution. This article does not contain any studies with human participants and/or specimens.

AUTHOR DISCLOSURE STATEMENT

No competing financial interests exist.

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